

Deep Drift: saved versions and how git works

A guide for Hasib. Keeps track of every saved version of the dive, and explains how to go back to one.
Updated 25 Sep 2026.

In one sentence: git is a program that takes a photo of the whole project each time we save (a *commit*), keeps every photo forever, and can put any old photo back. Before each realism change we have a safe copy to return to.

1. Where everything is

Git program	<code>C:\Program Files\Git\cmd\git.exe</code> (the one Windows uses) <code>C:\Program Files\Git\mingw64\bin\git.exe</code> (the same program, used inside the Git Bash window)
Git version	git version 2.54.0.windows.1
The project	<code>C:\Users\hasib\Downloads\deep-drift-project</code>
The history	<code>C:\Users\hasib\Downloads\deep-drift-project\.git</code> A hidden folder, about 9.4 MB. Every saved version lives inside it (packed and compressed). Never edit or delete it: deleting it deletes all history. Your normal files are not affected.
Settings for this project	<code>.git\config</code> . Name <code>Hasib</code> , email <code>hasibahmad.njupt@gmail.com</code> . These are set for this project only. Git on this computer has no global name or email.
Branches	<code>main</code> : released versions. <code>feature/v1.2-continuous-dive</code> , <code>feature/v1.3-solid-and-smooth</code> , <code>feature/v1.4-feel</code> : where each version was built (kept for the record). You are on <code>main</code> . See section 4.
Run it locally	In the project folder: <code>node tools/serve.mjs</code> , then open <code>http://localhost:8080</code> . Stop it with Ctrl+C.
Tags (named versions)	<code>v1.0</code> , <code>v1.1</code> , <code>v1.2</code> , <code>v1.3</code> , <code>v1.4</code>
Online copy	None. The history exists only on this computer. If the project folder is lost, the history is lost too. Putting a copy on GitHub (private) would fix that; ask when you want it.
Plain backup page	<code>backup\deep-drift-v1.0.html</code> . Double-click it to play the v1.0 dive. No git needed. (Any other version: see section 5.)
This guide	<code>docs\git-tracking.pdf</code> , made by <code>docs\make_git_tracking.py</code>
Special files	<code>.gitignore</code> : things git ignores (<code>node_modules</code> , screenshots <code>*.png</code> , <code>__pycache__</code>). <code>.gitattributes</code> : keeps line endings exactly as saved. Without it, Windows git would switch them when restoring, so an old version would not come back byte for byte.

2. Why the version is called v1.0, and how numbers go up

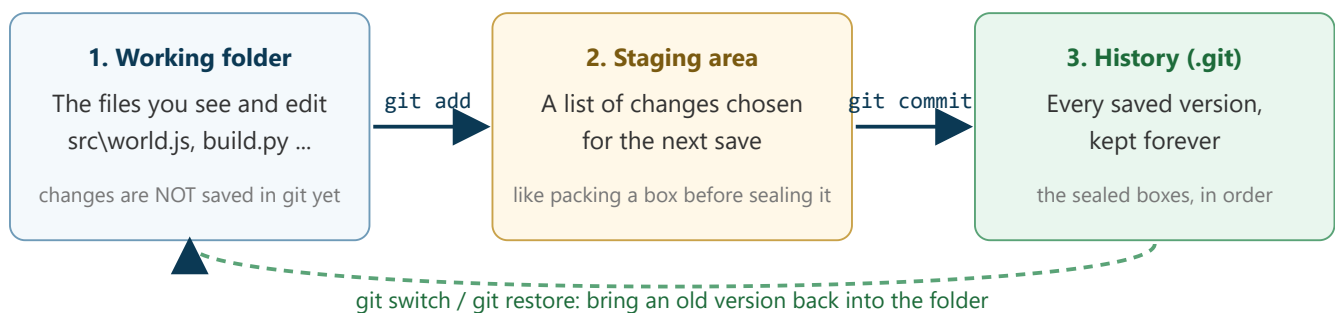
The project file `package.json` said `0.4.0` before git was set up. A number starting with 0 usually means "still being built". You chose to call the current dive **v1.0**: the first saved version. `package.json` was changed to `1.0.0` so everything agrees.

The rule from now on (**vMAJOR.MINOR**):

- **MINOR goes up by one** for each finished, tested change to the dive: v1.1 (the realism work), v1.2 (the continuous dive), and so on. After v1.9 comes v1.10, not v2.0.
- **MAJOR goes up** (v2.0) only for a very big change you decide on, for example a full redesign.
- **No new number** for commits that do not change the dive: this guide, test fixes, notes. They still get a row in the table, with "none" as the version.
- If a version is undone, its number is **not reused**. The next change takes the next number, so a number always means one thing.

3. How git works

The three places a change passes through



1. **Working folder.** The files as they are now. When code is edited, only this changes. Git notices, but nothing is saved yet.
2. **Staging area.** `git add` puts chosen changes here. It is a waiting list for the next save, so one save can hold one tidy change even if other files were also touched.
3. **History.** `git commit` seals everything on the waiting list into a new saved version inside `.git`. After that it is safe: later edits cannot damage it.

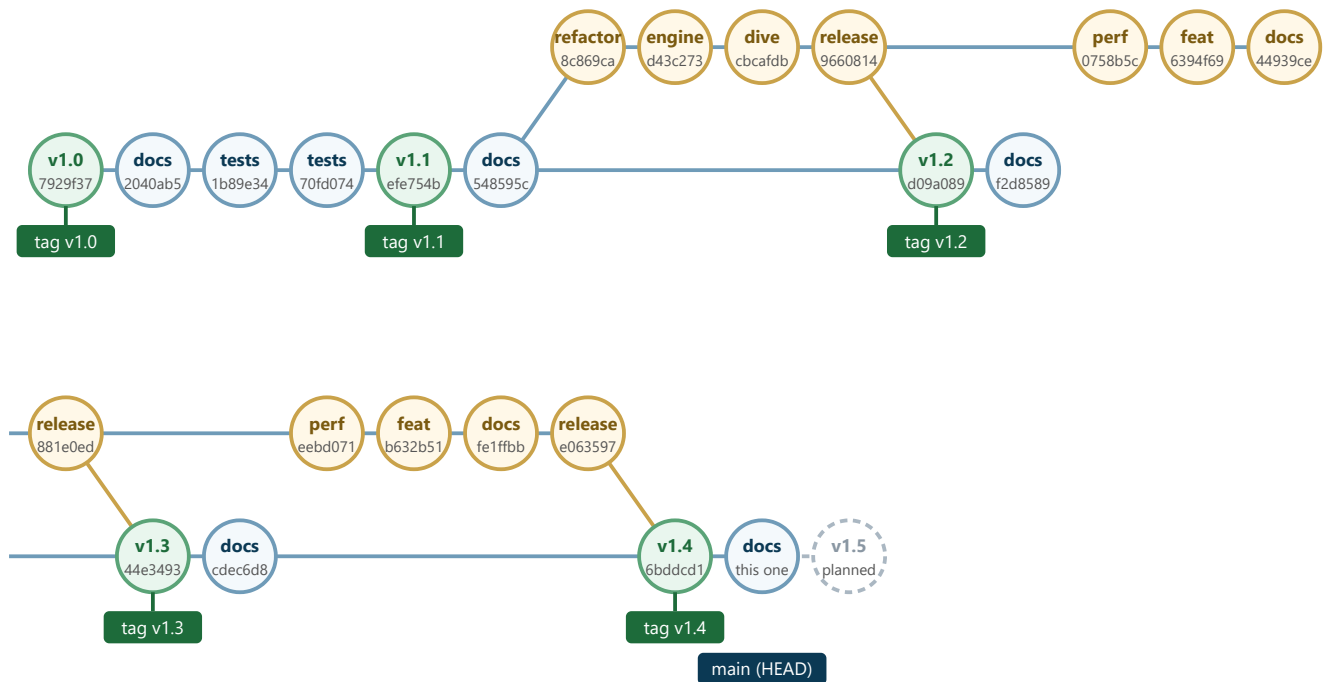
What a commit is

A commit is a complete snapshot of every tracked file at that moment, plus a message (what and why), the author, the date, and a pointer to the commit before it. Git stores only what changed, so saving often is cheap: the whole history today is about 9.4 MB.

Each commit gets an **ID** like `7929f37` (short form) or a 40-character long form. The ID is a fingerprint calculated from the commit's content. Change one letter anywhere and the ID changes completely. That is why an old version cannot be changed quietly. It is also why this file cannot show the ID of the commit that contains it: the ID is only known after the file is saved. That row is filled in at the next update.

History, branch, tag and HEAD

side branches: feature/... (one per version, where it was built)



Read left to right, row after row. Lower line: main (released versions). Upper line: side branches.

- **History** is a chain: every commit points back to its parent.
- **Branch**: a name that points at the newest commit of a line of work and moves forward with every commit. `main` holds the released versions; each new version is built on its own side branch first (the upper row).
- **Tag** (`v1.0`): a permanent name for one commit. It never moves. This is how "go back to v1.1" stays easy.
- **HEAD**: "where you are now", the version the working folder is based on. Normally HEAD is on `main`.

4. Branches, testing and releases (how each version is made)

Since v1.2 each version is built the way software teams do it (a simple "SDLC": build, test, release):

1. **Branch**. Start a side branch for the version, for example `git switch -c feature/v1.5-behaviour`. `main` is not touched, so the last released version is always safe there.
2. **Build in small steps**, each saved as its own commit on the branch (for v1.2: refactor, engine upgrade, the continuous dive). The code is split into small files, one job each, so every part can be found and changed later.
3. **Test locally** after every step: `node tools/serve.mjs` to play it at `http://localhost:8080`, and the test scripts (`tests\shot.py`, `ui.py`, `live.py`, `glide.py`, `solid.py`, `startup.py`, `life.py`).
4. **Show Hasib** before and after. Only if it looks better and nothing broke:
5. **Release**. Set the version in `package.json`, merge the branch into `main` with a merge commit (`git switch main` then `git merge --no-ff feature/...`), and tag it (`git tag -a v1.3 -m "..."`).
6. **Report**: after every fix, a PDF report of what is on the website and what to do next (`python docs/make_report.py`, into `docs/reports`).
7. **Record it**: add rows to `COMMITs` in `docs\make_git_tracking.py`, run it, and commit the new PDF.

8. **Publish** (later): `python tools\build.py` makes `dist\` with only the files a visitor needs; that folder goes to the host.

If a step looks worse or breaks something, it is not committed; the files are put back with `git restore`. If a problem is found only later, that one commit is undone with `git revert` (below), and the table shows it as "Undone". A whole branch can also simply be left unmerged: `main` never saw it.

5. How to look back and go back

Type these in **Git Bash** (Start menu, "Git Bash") or in PowerShell, after moving into the project folder: `cd C:\Users\hasib\Downloads\deep-drift-project`. Or ask Claude to do it.

<code>git log --oneline --decorate</code>	List every saved version, newest first, with IDs and tags.
<code>git log --oneline --graph --all</code>	The same, drawn with the branches and merges.
<code>git status</code>	Show which files changed since the last save. Safe, changes nothing.
<code>git diff</code>	Show the exact lines changed since the last save. Safe.
<code>git show v1.1:dist/index.html > old.html</code>	v1.0 and v1.1 were one single file: this saves one as <code>old.html</code> , without touching the project. Open it in a browser.
<code>git switch --detach v1.2</code> <code>node tools/serve.mjs</code> then <code>git switch main</code>	Play any version from v1.2 on: the folder turns into that version, the server runs it, and the last command comes back to the newest. Nothing is lost. (Save or undo current edits first; git will warn you.)
<code>git restore src/world.js</code>	Throw away unsaved edits to one file, back to the last commit.
<code>git revert 1a2b3c4</code>	Undo one commit by adding a new commit that does the opposite. Later changes are kept, and the history still shows what happened. The safe way to undo.

Careful: `git reset --hard v1.0` makes the project exactly v1.0 and **deletes unsaved edits and every commit after it** (tagged versions can still be recovered, others are hard to find). Use `git revert` or `git switch` instead, or ask Claude first.

6. The table of saved versions

Every commit, oldest first. The commit ID links a row to git: `git show 7929f37` shows everything in that commit.

#	Version	Commit ID	Date	Commit name	Detailed description	Status
1	v1.0	7929f37	24 Sep 2026, 18:22	Save the dive as it was before the realism work v1.0: save the dive as it was before the realism work	What changed: Nothing in the dive itself was changed. The whole project was put under git for the first time, so every later change can be undone. Files: - All of <code>src/</code>, <code>assets/</code>, <code>tests/</code>, <code>tools/</code>, <code>README.md</code>, <code>CLAUDE.md</code>, <code>build.py</code>, <code>.gitignore</code> <code>dist/index.html</code> : the built, playable page <code>package.json</code> : version changed from 0.4.0 to 1.0.0 - New: <code>backup/deep-drift-v1.0.html</code>, a plain copy of the page that opens without git - New: <code>.gitattributes</code>, keeps line endings exactly as saved State of the dive: Six places: reef, open ocean, midnight, vents, wreck, trench. One real 3D model (the barramundi fish); everything else is built by code. The weak spots listed in <code>CLAUDE.md</code> (hazy reef wall, dark wreck and trench, small deep animals, corals you can swim through, untested on phones) are all still present. Checks: All five scripts pass a syntax check, alone and joined. <code>dist/index.html</code> is identical to a fresh build. No screenshots yet: the test tools were not installed at this point.	Kept
2	none	2040ab5	24 Sep 2026, 18:25	Add this git tracking guide docs: add the git tracking guide	What changed: Added this PDF and the small script that makes it. The dive was not changed, so this commit gets no version number. Files: - New: <code>docs/git-tracking.pdf</code> (this file) - New: <code>docs/make_git_tracking.py</code> (writes this file; holds the table entries) State of the dive: Same as v1.0. Checks: PDF opened and read through.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
3	none	1b89e34	24 Sep 2026, 18:35	Make the tests work on Windows and apply the page policy for real tests: make the test tools work on Windows and apply the page policy for real	What changed: The screenshot tests were meant to run under a strict page policy like the real host, but the policy was never applied: it only matched an old file name, and the test browser cannot add it to file:// pages anyway. The page is now served from a made-up web address that the test script answers itself, so the policy really applies and anything not allowed is blocked. Screenshots go to <code>tests/out/</code> instead of <code>/tmp</code>, which does not exist on Windows. <code>build.py</code> now reads and writes UTF-8 and keeps line endings, so the built page is byte for byte the same. Files: <code>tests/shot.py</code>, <code>tests/ui.py</code>, <code>tests/live.py</code>, <code>tests/grid.py</code> <code>build.py</code>, <code>.gitignore</code> (ignores <code>tests/out/</code>) <code>CLAUDE.md</code>: how to install the test tools on Windows (the full npm install fails on <code>sharp</code>, which only the old fish-model tools need, so only <code>three.js</code> is fetched) State of the dive: Same as v1.0. First real screenshots taken: the "before" pictures in section 7. Checks: Screenshots at 9 dive moments under the strict policy, no page errors. The real fish model still loads under the policy.	Kept
4	none	70fd074	24 Sep 2026, 19:21	Tests: wait for the Begin button without eval tests: wait for the Begin button without eval	What changed: Once the strict policy really applied, one test helper (<code>wait_for_function</code>) broke, because it runs text as code and the policy forbids that. The tests now wait with <code>wait_for_selector</code>, which the policy allows. Found while testing v1.1. Files: <code>tests/shot.py</code>, <code>tests/ui.py</code>, <code>tests/live.py</code> State of the dive: Same as v1.0. Checks: All three test scripts run with no errors.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
5	v1.1	efe754b	24 Sep 2026, 19:21	A more realistic reef wall, wreck, trench and deep animals v1.1: a more realistic reef wall, wreck, trench and deep animals	What changed: Hasib asked for the four areas in one version. Reef wall: rock grain painted from three directions (never stretched) that catches the light like real bumps; ledges, deep cracks and lumpy rock; more colour from encrusting life; a new set of corals and sponges growing out of the wall (vase and tube sponges, sea fans, soft and plate corals, sea whips). Wreck: rebuilt as a 47 m steamship: keel, flat sides, raked bow, flat stern, plate seams, black and faded red paint, rotted plank deck with holes, deckhouse with windows and bridge, broken railings, davits, a snapped mast across the deck, hatches, a torn hole in the side; rusticles hanging with gravity (as on the Titanic, at the same depth), anchor chain, sponges and anemones on deck, a debris field with the fallen funnel. Trench: finer walls with rock layers and ledges, pale sediment on ledges, boulders at the foot of the walls, sea cucumbers on the floor; the dive ends about 7 m above the floor. Deep light: the deep water was set so murky you could see only about 8 m; it is now clear, as real deep water is, and a soft lamp moves with the diver. Deep animals: tube worms thicker and in dense clumps with red plumes; anglerfish dark with fangs and a blue-green lure, now in view about 9 s instead of half a second; dumbo octopus with webbed arms; snailfish passing close, plus a group feeding near the trench floor; shrimp shapes for vent shrimp and amphipods. Speed: only the current place is drawn, so the deep places cost about a quarter of v1.0; the reef costs the same as v1.0. Files: <code>src/core.js</code> : rock grain, bumps, rust and rock layers in <code>wet()</code> ; clearer deep water; lamp; softer torch <code>src/world.js</code> : reef wall and wall corals, new sponges, <code>buildShip</code> , <code>addRusticles</code> , debris field, trench walls, boulders, sea cucumbers, tube worms <code>src/life.js</code> : dumbo web, anglerfish teeth and colour, <code>shrimpGeo</code> , paler animals toned down <code>src/run.js</code> : animals that sink with the diver, dumbo and snailfish routes, lure light, trench path, drawing only the current place <code>dist/index.html</code> rebuilt; <code>package.json</code> 1.1.0; <code>CLAUDE.md</code> updated State of the dive: See the before and after pictures in section 7. Still to do: corals are not solid, phone	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
					testing, anglerfish teeth are too small to notice, trench rock layers are subtle. Checks: Syntax of all scripts; screenshots at 9 dive moments under the strict page policy; phone and desktop menu test; live play test (pause, restart, music). No page errors. Triangles per frame measured at 1300x800 against v1.0: reef 4.2 million (v1.0: 4.0), deep places about 1.0 million (v1.0: 4.0).	
6	none	548595c	24 Sep 2026, 19:23	Update this guide for v1.1 docs: update the tracking guide for v1.1	What changed: Added the four new commits to this table, before and after pictures (section 7), and a history diagram drawn from the real commits. Files: - docs/git-tracking.pdf , docs/make_git_tracking.py - New: docs/img/before-v1.1.jpg and docs/img/after-v1.1.jpg State of the dive: Same as v1.1. Checks: PDF opened and read through.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
7	none	8c869ca	24 Sep 2026, 21:14	Split the code into small modules; run the site locally refactor: split the code into small ES modules and run the site locally	What changed: Hasib asked for clean, separated code that can be found later, and for local testing before shipping. The five big script files (about 2,400 lines) became 55 small modules, one job each, in folders by area (engine, dive, world, life, diver, ui, audio, content). The page is a normal folder now: index.html, styles/, src/, and three.js and the fonts included in the project, so nothing is loaded from other websites. A small local server (<code>node tools/serve.mjs</code>) runs it at <code>http://localhost:8080</code>. The dive itself did not change: screenshots at 9 moments looked the same as v1.1. Files: • New folders <code>src/engine</code> , <code>src/dive</code> , <code>src/world</code> , <code>src/life</code> , <code>src/diver</code> , <code>src/ui</code> , <code>src/audio</code> , <code>src/content</code> , <code>src/util</code> • <code>index.html</code> , <code>styles/</code> , <code>vendor/three/</code> , <code>assets/fonts/</code> • <code>tools/serve.mjs</code> (local server), <code>tools/build.py</code> (makes <code>dist/</code> for publishing), <code>tools/vendor_three.py</code> • Removed: the old single-file build; <code>dist/</code> is no longer kept in git State of the dive: Same as v1.1. Checks: Screenshots at 9 moments, menus on phone and desktop, live play. A first Python server dropped requests on Windows and the page could hang while loading; the Node server loaded 6 of 6 times in about 2 s.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
8	none	d43c273	24 Sep 2026, 22:59	Upgrade the 3D engine to the current three.js engine: upgrade three.js from r128 (2021) to r186	What changed: three.js r128 (2021) became r186. This turns on modern colour management and physically based lights (light fades with distance, as in reality), the base for more realistic pictures. Strengths were converted so the dive looked the same. One bug found on the way: the new engine gives its fog colour already converted for the screen, which made every deep place flat bright blue; our fog now keeps its own colour. Files: • vendor/three/ (r186), tools/vendor_three.py • src/engine/ lights, environment, uniforms and wet.js ; small renames in several shaders State of the dive: Same as v1.1, on the new engine. Checks: Screenshots at 9 moments matched v1.1; menus and live play pass.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
9	none	cbcafdb	25 Sep 2026, 00:01	One continuous dive, gliding swim, life at real depths dive: one continuous 7-minute dive from the surface to the Challenger Deep	What changed: Hasib found the jumps between places jarring and wanted more to explore, kept real. One world: the dive now goes down the side of a volcanic island without any fade or depth jump: reef, the island cliff into the twilight and midnight zones, a vent terrace at ~1,600 m, the volcano slope, the abyssal plain with the wreck at ~5,000 m, the trench wall, and the floor of the Challenger Deep at 10,935 m. Swimming: no button. You drift toward wherever you look and are always sinking; looking up slows the sinking, looking down speeds it up; you can never swim back up, so you must aim early; a gentle current brings you back if you stray too far. Life at real depths: every animal only appears where it really lives (for example snailfish only between about 6,500 and 8,300 m). New: deep reef (sea fans, black corals, sponges), hatchetfish, vampire squid, gulper eel, grenadiers, tripod fish, bamboo corals, glass sponges, sea lilies, brittle stars, sea pigs, xenophyophores, manganese nodules, liquid CO2 droplets at the vents. Look: backscatter (specks lit in the torch beam, like real deep-sea footage), fog that melts into the background, realistic rock brightness. Honest page text: the places are moved closer together and the descent is sped up. Files: - New: <code>src/dive/depth.js</code>, <code>route.js</code>; <code>src/world/sites.js</code>, <code>terrain.js</code>, <code>benthos.js</code>, <code>culling.js</code>; <code>src/diver/glide.js</code>; <code>src/life/species-deep.js</code>, <code>cast-deep.js</code>; <code>tests/glide.py</code> - Changed: vents, wreck, trench, reef, corals, collision, camera, input, cast, extras, passers, facts, HUD, particles, page text, README, CLAUDE.md - Removed: <code>src/dive/timeline.js</code>, <code>src/dive/path.js</code> (the old six separate places) State of the dive: One continuous ~7 minute dive from the surface to 10,935 m. See the pictures in section 7. Checks: Screenshots down the whole dive under the strict page policy; menus on phone and desktop; live play; glide test (steers, never rises, sinks slower or faster, roaming limit). No page errors. Triangles per frame at 1300x800: reef about 4.7 million (v1.1: 4.2), open water 0.8 million, deep places 1.1 to 2.2 million.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
10	none	9660814	25 Sep 2026, 00:01	Set the version number to 1.2.0 release: set the version to 1.2.0	What changed: The last step on the branch before merging: <code>package.json</code> says 1.2.0. Files: <code>package.json</code> State of the dive: Same as the commit before. Checks: None needed.	Kept
11	v1.2	d09a089	25 Sep 2026, 00:01	Release v1.2: merge the branch into main v1.2: one continuous dive to the Challenger Deep, glide swimming, life at real depths	What changed: The merge commit that brings the four commits of <code>feature/v1.2-continuous-dive</code> into <code>main</code>, tagged <code>v1.2</code>. From here, <code>main</code> is the v1.2 dive. Files: Everything changed on the branch (the four rows above it). State of the dive: v1.2: one continuous dive to the Challenger Deep with gliding swim and life at real depths. Checks: All tests passed on the branch before merging.	Kept
12	none	f2d8589	25 Sep 2026, 00:05	Update this guide for v1.2 docs: update the tracking guide for v1.2	What changed: New rows, a history diagram that shows the branch and the merge, a new part explaining branches and releases (section 4), before and after pictures for v1.2, and the plan for v1.3 and later. Files: <code>docs/git-tracking.pdf</code>, <code>docs/make_git_tracking.py</code> - New: <code>docs/img/before-v1.2.jpg</code>, <code>docs/img/after-v1.2.jpg</code> State of the dive: Same as v1.2. Checks: PDF opened and read through.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
13	none	0758b5c	25 Sep 2026, 20:44	Open faster; no freezes at the start of the dive perf: open faster and stop freezes early in the dive	What changed: Hasib noticed lag when the page opens. Measured first (a new test, <code>tests/startup.py</code>), then fixed four causes. 1. Too much drawn: each kind of coral was one big batch covering the whole reef, so all 3.9 million reef triangles were drawn every frame, even behind you. Corals, sea-floor life, rocks, tube worms, nodules and sea cucumbers are now split into tiles, and only tiles on screen and within sight are drawn. Small corals shrink away smoothly with distance (when they are a few pixels across and already faded) instead of being drawn out to 250 m. Schools of fish are skipped when off screen or far away. The first view went from 4.7 million to 0.34 million triangles. 2. Drawing while loading: the page drew the half-built world between loading steps. Now it builds in steps with a progress message (for example: Loading the coral reef... 11%) and draws nothing until it is ready. 3. Shaders built during the dive: the first time a material is drawn, the browser builds its shader, a freeze each time. All are now built before Begin (<code>warmup.js</code>). 4. A hidden light: the anglerfish carries a light; hiding and showing it made three.js rebuild the shaders of everything on screen. The light now stays and just goes dark. Also: picture sharpness now drops quickly when frames are slow and rises back slowly (<code>quality.js</code>), and tiny point sprites keep their size when it changes (an old bug). Files: - New: <code>src/util/steps.js</code>, <code>src/engine/warmup.js</code>, <code>src/engine/quality.js</code>, <code>src/world/tiles.js</code>, <code>tests/startup.py</code> - Changed: <code>main.js</code>, <code>culling.js</code>, <code>corals.js</code>, <code>benthos.js</code>, <code>floor.js</code>, <code>vents.js</code>, <code>wreck.js</code>, <code>trench.js</code>, <code>school.js</code>, <code>cast.js</code>, <code>wet.js</code> (new option <code>shrink</code>), <code>lights.js</code>, particles, bubbles, effects State of the dive: Same dive, lighter to draw. See the chart in section 7. Checks: In the test browser: Begin ready after 2.5 s (v1.2: 5.4 s); 26 pictures in the first 6 s of the dive (v1.2: 9); no shader built after the warm-up. Screenshots looked the same as v1.2.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
14	none	6394f69	25 Sep 2026, 21:03	Nothing passes through the diver; nothing pops in or out feat: a solid world, animals that give way, nothing pops in or out	What changed: Hasib's two rules: you cannot pass through anything (animals move away, or you are stopped), and nothing appears or vanishes suddenly. Solid world: corals, sponges, bamboo corals, sea lilies, big rocks, tube worm clumps, the wreck's masts, railings, winches, funnel and boiler, and the whale bones are now solid (7,794 simple shapes kept in a grid, <code>solids.js</code>). The diver is a 0.45 m ball and slides around them. The wreck is checked with its real hull shape instead of one big box. Animals give way: each animal senses the diver's body and has its own comfort distance. Too close, it speeds up smoothly and swims aside. If it still touches you, a light animal is pushed aside, and a heavy one (whales, the whale shark, the manta) pushes you aside instead. School fish keep about 1.4 m away and part around you. No pops: passing animals start where you cannot see them (off screen or lost in the water), swim across, and leave only once out of sight again; their path never goes through rock. Glows (vents, anglerfish lure) fade with distance. Deep animals that sink with you slow down gently instead of stopping dead. Bugs found by the new test and fixed: animals more than 300 m away stopped moving for good (so after the dive restarted they were frozen and hidden); near the top of the reef wall the diver could be lifted 58 m in one frame; past the look-up limit the diver was pulled down 6 m in one frame; the reef collision stepped up to 2 m at once; the lifeboat arms on the wreck were never added (their code was inside a comment). Files: • New: <code>src/world/solids.js</code>, <code>tests/solid.py</code> • Changed: <code>collision.js</code>, <code>glide.js</code>, <code>touch.js</code>, <code>school.js</code>, <code>passers.js</code>, <code>motion.js</code>, <code>main.js</code>, <code>ship.js</code>, <code>wreck.js</code>, <code>corals.js</code>, <code>benthos.js</code>, <code>floor.js</code>, <code>vents.js</code>, <code>effects.js</code>, <code>debug.js</code> State of the dive: The world is solid, animals give way, nothing pops. See the whale skeleton pictures in section 7. Checks: <code>tests/solid.py</code> simulates the whole dive plus a reef run and a wreck run (10,900 steps): 0 times inside something solid, 0 animal or fish overlaps, 0 pops, 0 jumps. Direct contact: a whale shark pushed the diver 1.9 m aside; a jellyfish was nudged 0.7 m away. Glide, menu and live tests pass.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
15	none	44939ce	25 Sep 2026, 21:11	Describe v1.3 for the next person (and for Claude) docs: describe v1.3 in CLAUDE.md and README	What changed: How the solid world, tiles, warm-up, sharpness control, giving way and passers work; the rule never to hide a light; the two new tests; the plan moved on (animal behaviour is now v1.4, real models v1.5 and later). Files: CLAUDE.md , README.md State of the dive: Same as the commit before. Checks: Read through; no em dashes.	Kept
16	none	881e0ed	25 Sep 2026, 21:11	Set the version number to 1.3.0 release: set the version to 1.3.0	What changed: The last step on the branch before merging: package.json says 1.3.0. Files: package.json State of the dive: Same as the commit before. Checks: None needed.	Kept
17	v1.3	44e3493	25 Sep 2026, 21:11	Release v1.3: merge the branch into main v1.3: a solid world, smooth arrivals and a faster start	What changed: The merge commit that brings the four commits of feature/v1.3-solid-and-smooth into main , tagged v1.3 . Why 1.3 and not 2.0: it is the same dive, it just behaves more like the real thing and opens faster. A 2.0 would mean a big change in what the dive is (for example real 3D models everywhere). Files: Everything changed on the branch (the four rows above it). State of the dive: v1.3: nothing passes through the diver, nothing pops in or out, faster opening. Checks: All tests passed on the branch before merging.	Kept
18	none	cdec6d8	25 Sep 2026, 21:13	Update this guide for v1.3 docs: update the tracking guide for v1.3	What changed: New rows, the whale skeleton before and after pictures, the opening chart, and the plan for v1.4 and later. Files: - docs/git-tracking.pdf , docs/make_git_tracking.py - New: docs/img/solid-v1.3.jpg , docs/img/opening-v1.3.jpg State of the dive: Same as v1.3. Checks: PDF opened and read through.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
19	none	eebd071	25 Sep 2026, 23:52	A smooth start on Hasib's laptop; no blink at Begin perf: a smooth start on laptop graphics, no blink at Begin	What changed: Hasib saw the camera blink and lag at the beginning. This time it was measured on his laptop's own graphics card (Intel UHD, screen 1536x864 at 125%), not only the slow test graphics. The reef, the start of the dive, took 50 to 70 ms per picture; after a few seconds the automatic sharpness dropped, which cleared the picture for a frame (a blink) and caused a 170 to 200 ms hitch; and Begin cut the picture to black. Now: before Begin the page draws the reef a few times and picks the sharpness for about 30 ms per picture (<code>benchmark.js</code>); sharpness changes only after about a second of slow pictures and never between drawing and showing; no edge smoothing on scaled screens (a fifth of the cost there); cheaper rock grain; coral shapes share their corners; the warm-up copies shapes a few at a time so loading never freezes long; Begin fades the intro card away over the live picture. Files: • New: <code>src/engine/benchmark.js</code> , <code>src/lib/geometry-utils.js</code> • Changed: <code>renderer.js</code> , <code>quality.js</code> , <code>warmup.js</code> , <code>wet.js</code> , <code>corals.js</code> , <code>styles/main.css</code> , <code>tests/startup.py</code> (new options <code>--gpu</code> , <code>--screen</code> , <code>--freeze</code>) State of the dive: Smooth from the first second on Hasib's laptop, a little softer picture there. Checks: On the real graphics card at Hasib's size: first 8 to 12 s of the dive at 60 pictures a second, no hitch over 33 ms (v1.3: 200 ms), no sharpness change during the dive, longest loading freeze 0.37 s (v1.3: 1.85 s).	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
20	none	b632b51	25 Sep 2026, 23:52	Living pause, a new dive every time, smooth arrow keys, real fish speeds feat: hover while the sea lives on, a new dive every time, smooth keys, lively schools	What changed: Pause: it now stops only the diver; the sea has its own clock, so schools, sharks and jellyfish keep swimming, passing animals swim on, and new visitors still come. Variety: at Begin and every restart each scripted animal rolls its version: earlier or later, nearer or farther, either side, faster or slower, and about one dive in seven not there; circling animals and schools start anywhere and may circle the other way; the dolphins stay one pod. Every version is checked against rock with a new honest 0.5 m check; if none is clear the animal stays away. This found older mistakes: the three reef sharks circled through the reef wall and the manta met you inside the wall (it now glides over you). Keys: arrow keys speed up to a smooth turn and ease to a stop instead of 7-degree jumps. Fish: each tail beat moves a fish about 0.7 of its length (Bainbridge 1958), so small school fish now beat their tails about 8 times a second instead of 2; schools surge (speed up and slow down by about 40%); each fish faces its real movement. Files: - New: <code>src/life/variety.js</code> , <code>tests/life.py</code> - Changed: <code>main.js</code> , <code>state.js</code> , <code>camera.js</code> , <code>input.js</code> , <code>collision.js</code> (new <code>inRock</code> for animals), <code>motion.js</code> , <code>school.js</code> , <code>passers.js</code> , <code>touch.js</code> , <code>cast.js</code> , <code>cast-deep.js</code> , <code>extras.js</code> , <code>small-shapes.js</code> , <code>barramundi.js</code> , <code>hud.js</code> , <code>playback.js</code> , <code>debug.js</code> , <code>tests/solid.py</code> State of the dive: Pause hovers while the sea lives on; every dive differs; smooth keys; lively schools. Checks: <code>tests/life.py</code> : 3 s of pause, the diver moved 0.08 m (breathing) and all 8 schools and 12 animals nearby kept moving; 73 of 74 animals change between dives, 1 to 7 absent, 0 paths through rock; keys turn with no jump over 0.14 degrees per picture; a school's speed ranged from half to twice its average. <code>tests/solid.py</code> still 0 problems.	Kept

#	Version	Commit ID	Date	Commit name	Detailed description	Status
21	none	fe1ffbb	25 Sep 2026, 23:56	Describe v1.4; start the report after every fix docs: describe v1.4; a report after every fix; Claude's notes	What changed: Hasib's new rule: after every fix, a PDF report of what is on the website and what to do next, and a notes file Claude keeps for itself. <code>docs/make_report.py</code> writes <code>docs/reports/report-v1.4.pdf</code>; <code>docs/claude-notes.md</code> holds Claude's notes; the page style and PDF printing moved into <code>docs/pdf_tools.py</code>, shared by both PDF scripts. CLAUDE.md and README describe the v1.4 parts. Files: - New: <code>docs/make_report.py</code>, <code>docs/pdf_tools.py</code>, <code>docs/claude-notes.md</code>, <code>docs/reports/report-v1.4.pdf</code>, <code>docs/reports/img/</code> - Changed: <code>CLAUDE.md</code>, <code>README.md</code>, <code>docs/make_git_tracking.py</code> State of the dive: Same as the commit before. Checks: Report opened and read through; no em dashes.	Kept
22	none	e063597	25 Sep 2026, 23:56	Set the version number to 1.4.0 release: set the version to 1.4.0	What changed: The last step on the branch before merging: <code>package.json</code> says 1.4.0. Files: <code>package.json</code> State of the dive: Same as the commit before. Checks: None needed.	Kept
23	v1.4	6bddcd1	25 Sep 2026, 23:56	Release v1.4: merge the branch into main v1.4: a smooth start, a living pause, a new dive every time	What changed: The merge commit that brings the four commits of <code>feature/v1.4-feel</code> into <code>main</code>, tagged <code>v1.4</code>. The planned animal behaviour moves to v1.5, because these fixes came first (a number always means one thing). Files: Everything changed on the branch (the four rows above it). State of the dive: v1.4: smooth start, living pause, a new dive every time, smooth keys, real fish speeds. Checks: All tests passed on the branch before merging.	Kept
24	none	Added at the next update (see section 3)	this commit	Update this guide for v1.4 docs: update the tracking guide for v1.4	What changed: New rows, the start-of-dive chart, and the plan for v1.5 and later. Files: <code>docs/git-tracking.pdf</code>, <code>docs/make_git_tracking.py</code> State of the dive: Same as v1.4. Checks: PDF opened and read through.	Kept

Planned (not saved yet)

v1.1 (the realism work), v1.2 (the continuous dive), v1.3 (the solid world and faster start) and v1.4 (smooth start, living pause, variety) are done. After every fix there is also a report in docs/reports. What is left is below. Rows move into the table above when they are committed.

Version	Change	What it will do
v1.5	Animal behaviour	Curious and startled school fish, predators chasing schools, hidden animals (day octopus with ink, flounder, scorpionfish, garden eels), deep-sea light displays, feeding at the vents and the whale skeleton, animals steering around corals.
v1.6+	Real 3D models	In small batches from Sketchfab, each with Hasib's OK on its licence and a credit in the page.
none	Lighter reef	Simpler far corals, so laptops like Hasib's can draw the reef at full sharpness.
none	Phone check	Try the dive on a real phone: speed, motion look (gyro), and the look-to-swim glide. Needs Hasib's phone.
none	Publish	First public version on GitHub Pages (free), after local testing. Needs a GitHub account.

7. Before and after pictures

Taken by the test script with the same settings, at the same moments of the dive. The test window counts as a small screen, so these show the lighter phone-level detail.

Start of the dive on Hasib's laptop (Intel UHD graphics, 1536x864 at 125%): v1.3 (grey) and v1.4 (blue)

Longest hitch in the first 10 to 12 s of the dive (shorter is better)



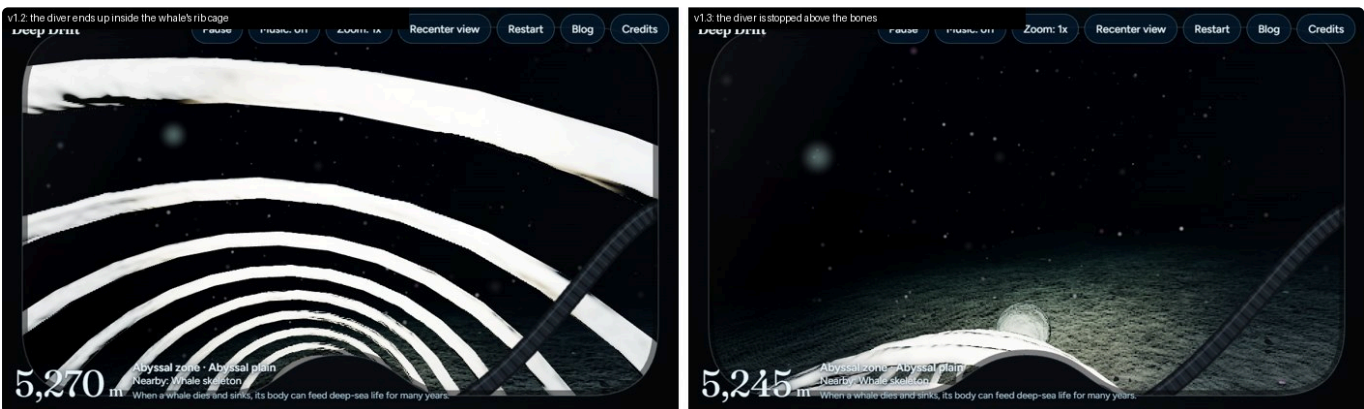
Time for one picture of the busy reef (full sharpness 1.25) (shorter is better)



Loading screen: longest freeze (shorter is better)



v1.4, the start of the dive on Hasib's laptop (Intel UHD graphics, 1536x864 at 125%): v1.3 and v1.4.



v1.3, solid world. The diver is sent straight into the whale skeleton on the abyssal plain. Left, v1.2: the camera ends up inside the rib cage. Right, v1.3: the diver is stopped just above the bones.

Opening the page: v1.2 (grey) and v1.3 (blue), measured in the test browser (software graphics, slower than a real computer)

Triangles drawn in the first view (fewer is lighter)



Time until Begin can be pressed (shorter is better)



Pictures drawn in the first 6 s of the dive (more is smoother)



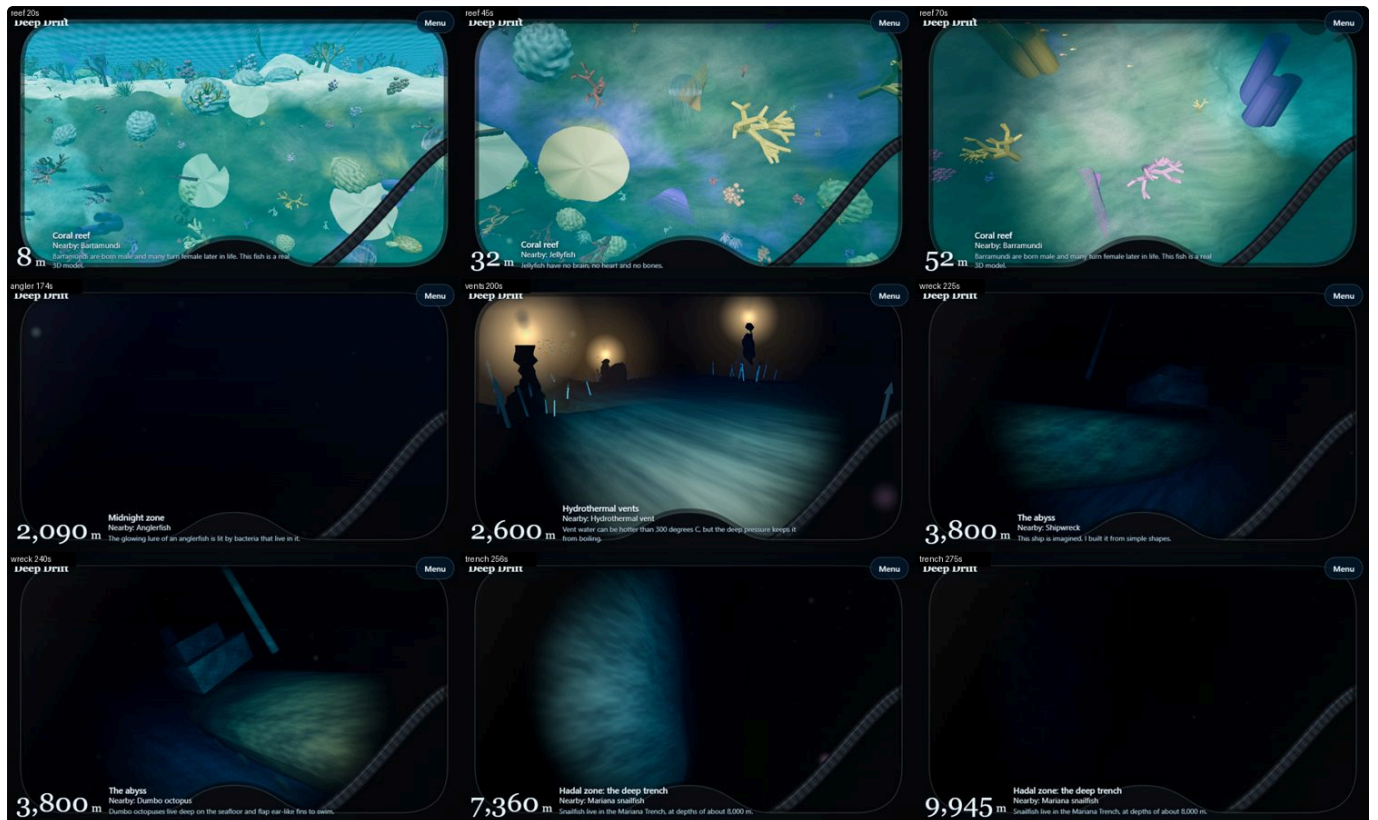
v1.3, opening the page, measured in the test browser (software graphics, so all numbers are slower than on a real computer; compare the bars, not the numbers).



v1.1 (before v1.2): six separate places, with a fade to black between the vents, the wreck and the trench.



v1.2: one continuous dive. Left to right, top to bottom: reef 11 m and 54 m, reef drop-off 155 m, twilight 560 m (hatchetfish), midnight 1,120 m (vampire squid) and 1,450 m (anglerfish), vents 1,580 m, volcano slope 2,270 and 3,440 m, abyssal plain 4,840 m, wreck 4,860 m, whale skeleton 5,185 m, trench 6,300 and 8,275 m, Challenger Deep 10,890 m.



Before (v1.0). Top: reef at 8, 32 and 52 m. Middle: midnight zone (the anglerfish should be here), vents, wreck. Bottom: wreck, trench, trench.



After (v1.1), at the same moments of the dive (the anglerfish and trench shots are a few seconds later, when the animals pass).

8. Words used in this guide

Repository (repo)	The project folder plus its <code>.git</code> history.
Commit	One saved snapshot of the whole project, with a message.
Commit ID (hash)	The fingerprint name of a commit, like <code>7929f37</code> .
Stage / staging area	The waiting list of changes for the next commit.
Branch	A moving name for the newest commit of a line of work: <code>main</code> for releases, a side branch per new version.
Merge	Bringing a side branch into <code>main</code> . The merge commit has two parents: the last commit on each line.
Release	A tested version merged into <code>main</code> and given a tag, like <code>v1.2</code> .
Tag	A fixed name for one commit, used for version numbers.
HEAD	The commit the working folder is currently based on.
Revert	Undo a commit by adding its opposite as a new commit.
Remote	A copy of the repository somewhere else, like GitHub. We have none yet.